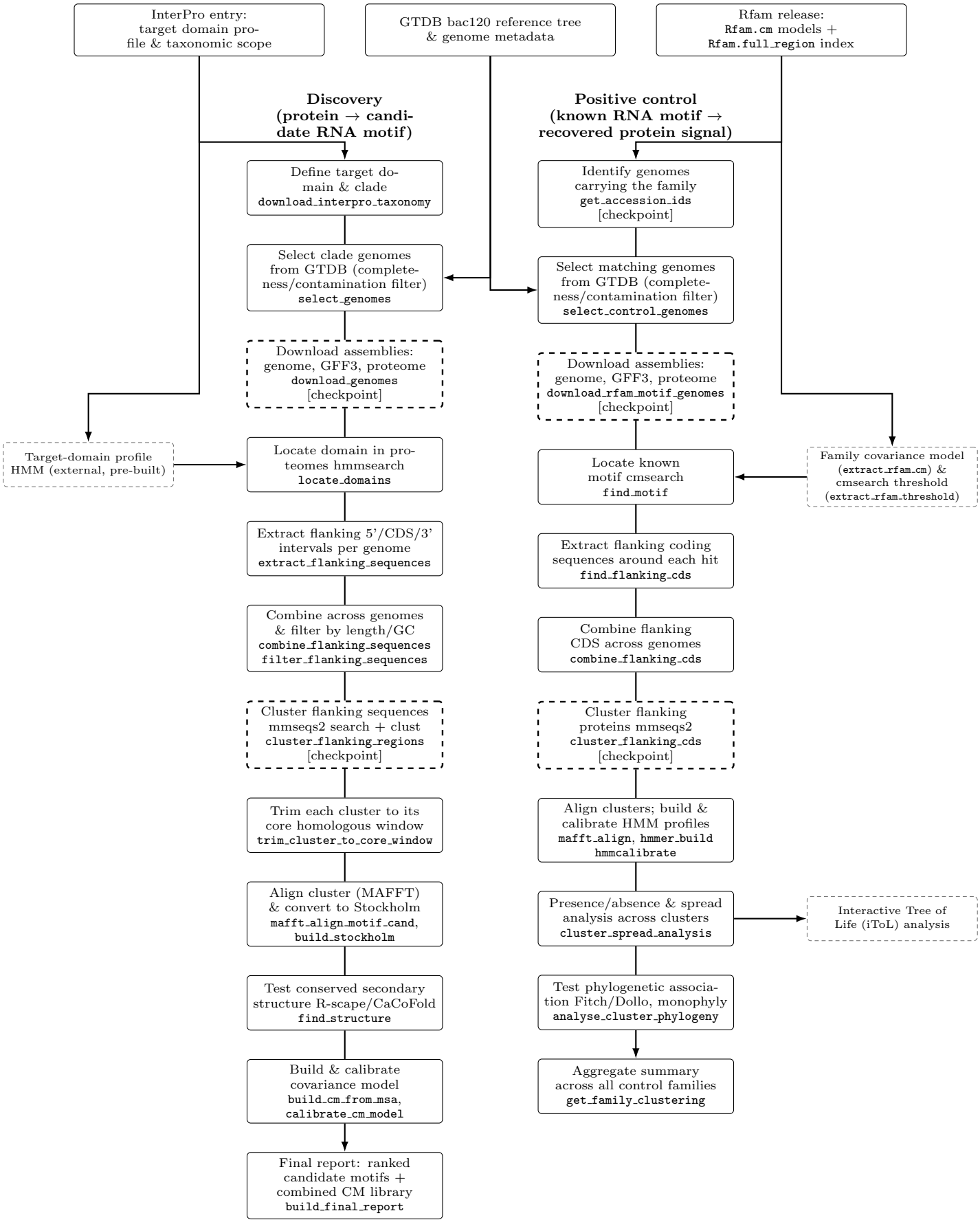

InterPro entry: target domain profile & taxonomic scope
GTDB bac120 reference tree & genome metadata
Rfam release: Rfam.cm models + Rfam.full_region index
Discovery (protein → candidate RNA motif)
Positive control (known RNA motif → recovered protein signal)
Define target domain & clade download_interpro_taxonomy
Select clade genomes from GTDB (completeness/contamination filter) select_genomes
Download assemblies: genome, GFF3, proteome download_genomes [checkpoint]
Target-domain profile HMM (external, pre-built)
Locate domain in proteomes hmmsearch locate_domains
Extract flanking 5'/CDS/3' intervals per genome extract_flanking_sequences
Combine across genomes & filter by length/GC combine_flanking_sequences filter_flanking_sequences
Cluster flanking sequences mmseqs2 search + clust cluster_flanking_regions [checkpoint]
Trim each cluster to its core homologous window trim_cluster_to_core_window
Align cluster (MAFFT) & convert to Stockholm mafft_align_motif_cand, build_stockholm
Test conserved secondary structure R-scape/CaCoFold find_structure
Build & calibrate covariance model build_cm_from_msa, calibrate_cm_model
Final report: ranked candidate motifs + combined CM library build_final_report
Identify genomes carrying the family get_accession_ids [checkpoint]
Select matching genomes from GTDB (completeness/contamination filter) select_control_genomes
Download assemblies: genome, GFF3, proteome download_rfam_motif_genomes [checkpoint]
Family covariance model (extract_rfam_cm) & cmsearch threshold (extract_rfam_threshold)
Locate known motif cmsearch find_motif
Extract flanking coding sequences around each hit find_flanking_cds
Combine flanking CDS across genomes combine_flanking_cds
Cluster flanking proteins mmseqs2 cluster_flanking_cds [checkpoint]
Align clusters; build & calibrate HMM profiles mafft_align, hmmer_build hmmcalibrate
Presence/absence & spread analysis across clusters cluster_spread_analysis
Interactive Tree of Life (iToL) analysis
Test phylogenetic association Fitch/Dollo, monophyly analyse_cluster_phylogeny
Aggregate summary across all control families get_family_clustering
Legend
External / static input file
Snakemake checkpoint (dynamic re-evaluation)